Causal Machine Learning Project

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1 Introduction

Some stuff to be introduced.

1.1 Research question

Research question

How does the predictive strength of covariates that are only outcome-relevant affect the relative performance of Dragonnet and TARNet in estimating the Average Treatment Effect (ATE)?

1.2 Hypothesis

Research question

As the predictive strength of outcome-only covariates increases, Dragonnet's relative advantage over TARNet in ATE estimation may diminish, because accurate modeling of said covariates becomes increasingly valuable.

2 Statistical Foundations

2.1 Causal identification

- Define the ATE using potential outcomes: $\tau = \mathbb{E}[Y(1) - Y(0)]$.
- Introduce identification assumptions.

2.2 Propensity-score sufficiency

- Introduce this result from (Rosenbaum and Rubin 1983): $e(X) = P(T = 1|X)$.
- Explain why conditioning on the true propensity score is sufficient for confounding adjustment under the identification assumptions.
- Make sure to distinguish between: a) confounders, which influence treatment and outcome; b) outcome-only predictors, which influence outcomes but not treatment assignment. > A variable can be unnecessary for identification but useful for improving statistical efficiency.

3 Neural Networks for Causal Estimation

- Compare the architectures of TARNet and Dragonnet. Explain that both use a shared representation and two outcome heads, with Dragonnet adding a third head to predict treatment assignment.
- Present training objectives.

4 Designing the Experiment

4.1 Data-Generating Process

- Simulate observational data with $X=(X_C, X_O), where X_C denotes confounders and X_O denotes outcome-only covariates.
- Treatment assignment:

$$T_i \sim \text{Bernoulli}\big(e(X_{Ci})\big), \quad \text{where}$$

$$e(X_{Ci}) = \sigma\big(f(X_{Ci})\big).$$

- Potential outcomes:

$$Y_i(0) = g(X_{Ci}) + \gamma h(X_{Oi}) + \epsilon_i \quad \text{and}$$

$$Y_i(1) = Y_i(0) + \tau, \quad \tau = 2.$$

- The parameter γ controls the predictive strength of outcome-only covariates.

4.2 Simulation Scenarios

- Hold the number of covariates, sample size, confounding mechanism, and noise variance constant; vary only γ:

5 Results

A silly block just to see how the code appears:

```
ducks = pd.DataFrame({
    "lamp": ["Still", "Still", "Swiveled", "Swiveled"],
    "seconds": [12, 14, 9, 11],
})

mean_times = ducks.groupby("lamp")["seconds"].mean()
for lamp, seconds in mean_times.items():
    print(f"{lamp}: {seconds:.1f} seconds")
```

```
Still: 13.0 seconds
Swiveled: 10.0 seconds
```

6 Discussion

Some stuff to be discussed.

References

Rosenbaum, Paul R., and Donald B. Rubin. 1983. “The Central Role of the Propensity Score in Observational Studies for Causal Effects.” *Biometrika* 70 (1): 41–55. https://doi.org/10.1093/biomet/70.1.41.